

Qatar-2b

R-1361

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_210422 · created 2026-07-31T23:38:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459326.7445 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.205 +/- 0.012
Transit depth (Rp/Rs)^2	4.21 +/- 0.51 [%]
Orbital Inclination (inc)	89.92 +/- 1.42
Airmass coefficient 1 (a1)	0.942 +/- 0.022
Airmass coefficient 2 (a2)	0.039 +/- 0.014
Scatter in the residuals of the lightcurve fit is	2.34 %
Best Comparison Star	#2 - [122, 24]
Optimal Method	PSF photometry
Transit Duration (day)	0.0792 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.205 $\pm$ 0.012 vs 0.16327 (deviation 3.48 σ)
Duration fit vs published	0.0792 d vs 0.0767 d (deviation 1.95 σ)
Mid-time O-C	2.5 min
Depth SNR	17.1
Residual scatter	2.34 %

Light curve

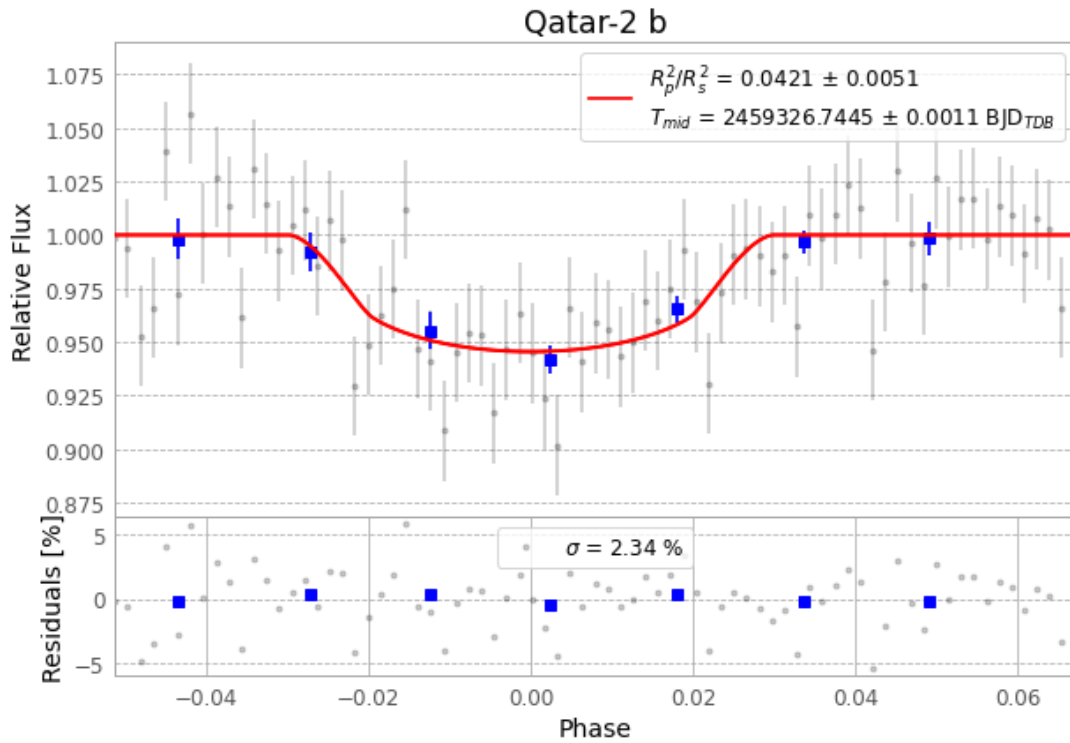


Figure 1 — FinalLightCurve_Qatar-2 b_2021-04-21.png (SHA-256 31b62a31dc3072f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [553, 34], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 53a0cb2f979f4229...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), Qatar-2210422040312.FITS → Qatar-2210422075112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-210422.log (9fddf102049b...), FinalLightCurve_Qatar-2 b_2021-04-21.png (31b62a31dc30...), FinalLightCurve_Qatar-2 b_2021-04-21.csv (6993a40d4cd9...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)